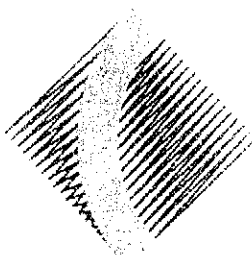


Name: _____
Class: 12MTZ1
Teacher: MR FARDOULY

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2006 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS EXTENSION 2

*Time allowed - 3 HOURS
(Plus 5 minutes' reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 8.

*****Each page must show your name and your class. *****

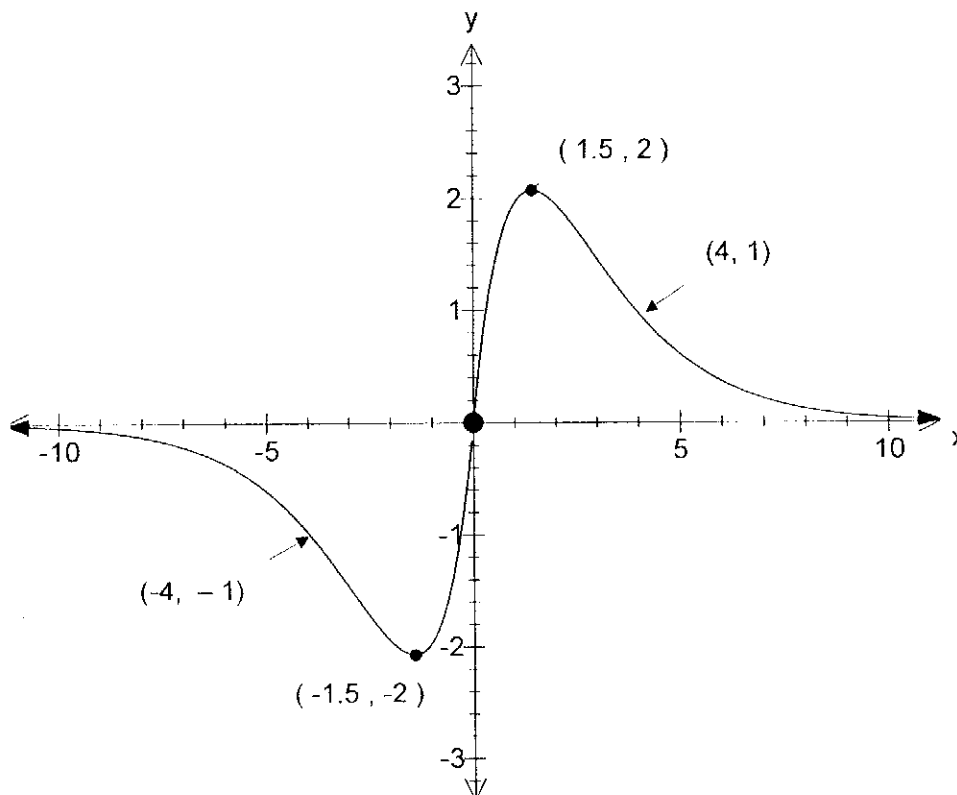
Question 1**Marks**

- (a) Find $\int x \sin(x^2 + 3) dx$ 2
- (b) Find $\int \frac{dt}{\sqrt{7 + 6t - t^2}}$. 2
- (c) Using the substitution $t = \tan \frac{\theta}{2}$, find $\int \frac{2}{4 + 3 \sin \theta} d\theta$. 3
- (d) (i) Show that $\frac{1}{(x^2 + 3)(x^2 + 1)} = \frac{1}{2} \left[\frac{1}{x^2 + 1} - \frac{1}{x^2 + 3} \right]$ 2
- (ii) Hence evaluate $\int_0^1 \frac{dx}{(x^2 + 3)(x^2 + 1)}$ 2
- (e) If $I_m = \int_0^k (k^2 - x^2)^m dx$, for $m \geq 1$, show that $I_m = \frac{2k^2 m}{2m + 1} \cdot I_{m-1}$. 4
- (Hint: $\frac{x^2}{k^2 - x^2} = \frac{k^2}{k^2 - x^2} - 1$)

Question 2 (Begin a new page)

- (a) Simplify $\frac{(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12})(\cos \frac{3\pi}{12} + i \sin \frac{3\pi}{12})}{\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}}$ 2
- (b) If $z = \sqrt{3} + i$, find z^4 , writing the answer in modulus-argument form. 2
- (c) The equation $z^2 - (a + bi)z - 6i = 0$, where a and b are real, has roots α and β such that $\alpha^2 + \beta^2 = 5$.
- (i) Show that $\alpha^2 - \beta^2 = 5$ and $\alpha\beta = -6$. 2
- (ii) Hence find the values of a and b . 2

- (a) The diagram shows the graph of $y = f(x)$



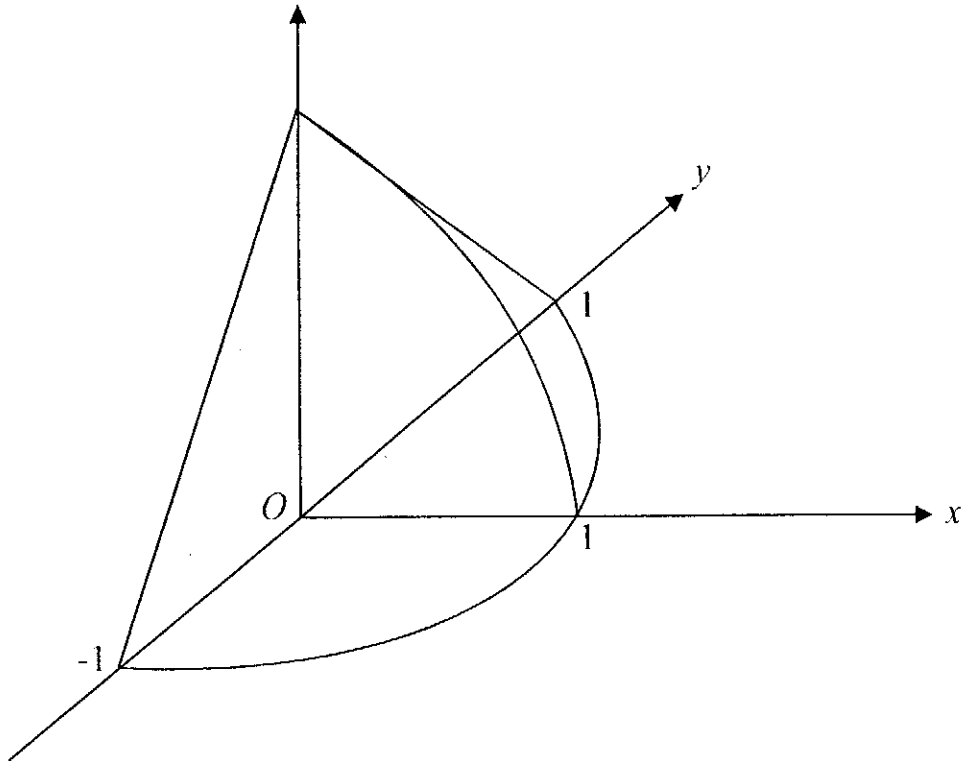
Draw separate sketches of the following:

- | | | |
|-------|--|---|
| (i) | $y = \frac{1}{f(x)}$ | 2 |
| (ii) | $y = [f(x)]^2$ | 2 |
| (iii) | $y = f'(x)$ | 2 |
| (iv) | $y = \int f(x) dx$, if $x = 0$ when $y = 0$ | 2 |
| (v) | $y = x + f(x)$ | 2 |

(Question 3 continued)

(b)

Marks



The base of a solid is the semi-circular region in the $x - y$ plane with the straight edge running from the point $(0, -1)$ to the point $(0, 1)$ and the point $(1, 0)$ on the curved edge of the semicircle. Each cross-section perpendicular to the x -axis is an isosceles triangle with each of the two equal sidelengths three quarters the length of the third side.

- (i) Show that the area of the triangular cross-section at $x = a$ is $\frac{\sqrt{5}}{2}(1 - a^2)$. 3
- (ii) Find the volume of the solid. 2

Question 4 (Begin a new page)

- (a) If p , q and r are the roots of the equation $x^3 + 4x^2 - 3x + 1 = 0$, 2
find the equation whose roots are $\frac{1}{p}$, $\frac{1}{q}$ and $\frac{1}{r}$.

- (b) (i) Let k be a zero of the polynomial $F(x)$ and also of its derivative $F'(x)$.
Prove that k is a zero of $F(x)$ of multiplicity at least 2. 3
- (ii) Show that $y = 1$ is a double root of the equation $y^{2t} - ty^{t+1} = 1 - ty^{t-1}$, where t is a positive integer. 2
- (c) (i) Determine the complex roots of $z^5 = 1$. 2
- (ii) Hence factorise $z^5 - 1$ over the
- (α) Complex field 2
- (β) Real field. 2
- (d) Show that for the complex number $z = \frac{1-t^2+2it}{1+t^2}$, $|z| = 1$ for all values of t . 2

Question 5 (Begin a new page)

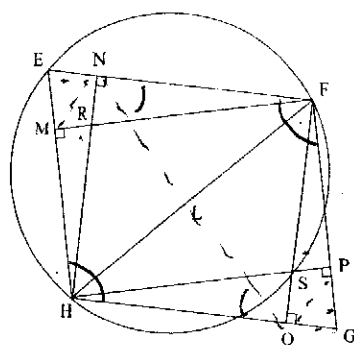
- (a) (i) The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $0 < a < b$, has eccentricity e . S is the focus of the hyperbola on the positive x -axis and the line through S perpendicular to the x -axis intersects the hyperbola at P and Q .
- (i) Show that $PQ = \frac{2b^2}{a}$. 2
- (ii) If P and Q have coordinates $(9, 24)$ and $(9, -24)$ respectively, show that $a = 3$ and $b = 6\sqrt{2}$. 3
- (iii) For these values of a and b , sketch the graph of the hyperbola showing clearly the x -intercepts, the coordinates of the foci, and the equations of the directrices and asymptotes. 4

- (b) An ellipse can be described as the locus of a point moving so that the sum of its distances from two fixed points (foci) is constant.
- (i) If the two fixed points are A(-4, 0) and B(4, 0) and the sum of the distances of $P(x, y)$ from these points is 10 units, show that the equation of the ellipse is $\frac{x^2}{25} + \frac{y^2}{9} = 1$. 2
- (ii) Show that the ellipse can be represented parametrically by the equations $x = 5 \cos \theta$ and $y = 3 \sin \theta$, and find the equation of the tangent to the ellipse at the point where $\theta = \frac{\pi}{6}$. 4

Question 6 (Begin a new page)

- (a) A body is projected vertically upwards from the ground with initial velocity v_0 in a medium that produces a resistance force per unit mass of kv^2 , where v is the velocity and k is a positive constant. Taking acceleration due to gravity as $g \text{ ms}^{-2}$,
- (i) Prove that the maximum height H of the body above the ground is $H = \frac{1}{2k} \log_e \left(1 + \frac{kv_0^2}{g} \right)$. 4
- (ii) Show that in order to double the maximum height reached, the initial velocity must be increased by a factor of $\left(e^{2kH} + 1 \right)^{\frac{1}{2}}$. 4
- (b) A body of mass m kg is moving in a horizontal straight line. At time t seconds it has displacement x metres from a fixed point O on the line and velocity $v \text{ ms}^{-1}$ and acceleration $a \text{ ms}^{-2}$. If the body is initially at O with velocity $V \text{ ms}^{-1}$, and $a = -\frac{1}{10} \sqrt{v} (1 + \sqrt{v}) \text{ ms}^{-2}$,
- (i) Show that $t = 20 \log_e \left(\frac{1 + \sqrt{V}}{1 + \sqrt{v}} \right)$ 3
- (ii) Find the distance travelled before the body comes to rest. 4

- (a) The vertices E, F and H of the parallelogram EFGH lie on a circle. L is the midpoint of the diagonal FH. R is the point of intersection of the altitudes HN and FM in the triangle EFH. S is the point of intersection of the altitudes HP and FQ in the triangle FGH. If S lies on the circle,



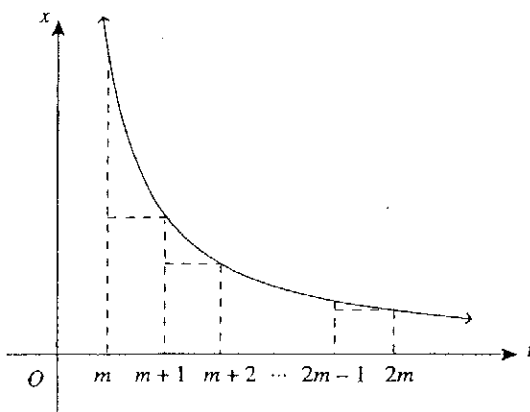
- (i) Prove that the points R, L and S are collinear. 4
- (ii) Show that the hexagon MNFPQH is cyclic. 1

- (b) (i) Prove that $\frac{1}{2p+1} + \frac{1}{2p+2} > \frac{1}{p+1}$, for all $p > 0$ 2

- (ii) Prove the following statement by mathematical induction 4

$$\frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{2m} \geq \frac{37}{60}, \text{ for } m \geq 3.$$

- (iii) The diagram below shows the graph of $x = \frac{1}{t}$, for $t > 0$.



- (α) By comparing areas, show that $\int_m^{m+1} \frac{1}{t} dt > \frac{1}{m+1}$. 1

- (β) Hence, without using a calculator, show that $\log_e 2 > \frac{37}{60}$. 3

Question 8 (Begin a new page)**Marks**

- (a) Given a , b and c are three non negative numbers, show that the arithmetic mean is greater than or equal to the geometric mean. 3
- (b) Polynomial $P(x)$ gives remainders -2 and 1 when divided by $2x - 1$ and $x - 2$ respectively. What is the remainder when $P(x)$ is divided by $2x^2 - 5x + 2$? 3
- (c) The equation $ax^4 + bx^3 + cx^2 + dx + e = 0$ has a quadruple root α .
- (i) Find α in terms of a and b . 2
- (ii) Hence, show that $\left(1 + \frac{b}{4a}\right)^4 = \frac{a + b + c + d + e}{a}$. 2
- (d) (i) If $n = -1$, show that $\int_1^e x^n \log x \, dx = \frac{1}{2}$ 2
- (ii) If $n \neq -1$, show that $\int_1^e x^n \log x \, dx = \frac{ne^{n+1} + 1}{(n+1)^2}$ 3

END OF TEST

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

AP4 EXTENSION 2 SOLUTIONS 2006

Question 1

(a) Let $u = x^2 + 3$

$du = 2x dx$

$\int u \sin(x^2 + 3) dx =$

$\frac{1}{2} \int \sin u du$ ✓

$= -\frac{1}{2} \cos u + c$

$= -\frac{1}{2} \cos(x^2 + 3) + c$ ✓

(b) $\int \frac{dt}{\sqrt{16 - (t-3)^2}}$ ✓

$= \sin^{-1}\left(\frac{t-3}{4}\right) + c$ ✓

(c) $\frac{d\theta}{d\phi} = \frac{1}{2} \sec^2 \frac{\theta}{2}$

$d\theta = \frac{2d\phi}{1+t^2}$ ✓

$\therefore \int \frac{2}{4+3\sin\theta} d\theta =$

$\int \frac{2}{4+3\left(\frac{2t}{1+t^2}\right)} \cdot \frac{2dt}{1+t^2}$

$\int \frac{2}{2t^2+3t+2} dt$ ✓

$= \int \frac{1}{\left(t+\frac{3}{4}\right)^2 + \frac{7}{16}} dt$

$= \int \frac{1}{\left(\frac{\sqrt{7}}{4}\right)^2 + \left(t+\frac{3}{4}\right)^2} dt$

$= \frac{4}{\sqrt{7}} \tan^{-1}\left(\frac{4t+3}{\sqrt{7}}\right) + c$ ✓

(d) (i) Let $\frac{1}{(x^2+3)(x^2+1)} = \frac{A}{x^2+3} + \frac{B}{x^2+1}$

$1 = A(x^2+1) + B(x^2+3)$

Let $x^2 = -1, \therefore 1 = 2B$

$B = \frac{1}{2}$

Let $x^2 = -3, \therefore 1 = A \cdot 2$

$\therefore A = \frac{1}{2}$

$\frac{1}{(x^2+3)(x^2+1)} = \frac{1}{2} \left[\frac{1}{x^2+1} - \frac{1}{x^2+3} \right]$ ✓

OR

{ Show RHS expanded + simplified }
equals LHS

(ii) $\int_0^1 \frac{dx}{(x^2+3)(x^2+1)} =$

$\frac{1}{2} \int_0^1 \frac{1}{x^2+1} - \frac{1}{x^2+3} dx$

$= \frac{1}{2} \left[\tan^{-1} x - \frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} \right]_0^1$ ✓

$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6} \right]$

$= \frac{\pi}{4} \left[\frac{1}{2} - \frac{1}{3\sqrt{3}} \right]$ ✓

(e) $I_m = \int_0^k (k^2 - x^2)^m \cdot 1 dx$

Let $u = (k^2 - x^2)^m$ ✓

$du = m(k^2 - x^2)^{m-1} \cdot -2x$ ✓

$\therefore I_m = \left[x(k^2 - x^2)^m \right]_0^k + \int_0^k 2mx(k^2 - x^2)^{m-1} dx$ ✓

$= \left[k(k^2 - k^2)^m - 0 \right] + 2m \int_0^k x^2 (k^2 - x^2)^{m-1} dx$

$= 2m \int_0^k \frac{x^2 (k^2 - x^2)^m}{k^2 - x^2} dx$

$= 2m \int_0^k (k^2 - x^2)^{m-1} (k^2 - x^2) dx$ ✓

$= 2mk^2 \int_0^k (k^2 - x^2)^{m-1} dx - 2m \int_0^k (k^2 - x^2)^m dx$

$\therefore I_m = 2k^2 I_{m-1} - 2m I_m$ ✓

$I_m (1 + 2m) = 2k^2 I_{m-1}$

$I_m = \frac{2k^2}{2m+1} I_{m-1}$ ✓

Question 2

$$(a) \cos\left(\frac{5\pi}{12} + \frac{3\pi}{12} - \frac{2\pi}{3}\right) + i\sin\left(\frac{5\pi}{12} + \frac{3\pi}{12} - \frac{2\pi}{3}\right)$$

$$= \cos\pi + i\sin\pi$$

$$= -1$$

$$(b) z = \sqrt{3} + i$$

$$= 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)$$

$$= 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$$

$$= 2e^{i\frac{\pi}{6}}$$

$$\therefore z^4 = 2^4 e^{i4\frac{\pi}{6}}$$

$$= 16 e^{i\frac{2\pi}{3}}$$

$$(c) (i) \alpha + \beta = a + bi$$

$$\alpha\beta = -b$$

$$(\alpha + \beta)^2 = \alpha^2 + \beta^2 + 2\alpha\beta$$

$$\therefore (a + bi)^2 = 5 - 12i$$

$$a^2 - b^2 + 2abi = 5 - 12i$$

equating real + imaginary parts

$$\left. \begin{aligned} a^2 - b^2 &= 5 \\ ab &= -6 \end{aligned} \right\}$$

$$(ii) \alpha^2 \cdot \beta^2 = 5$$

$$\therefore \alpha^4 \cdot \beta^4 = 5\alpha^2$$

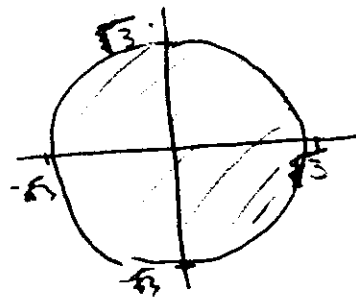
$$a^4 - 5a^2 - 36 = 0$$

$$(a^2 - 9)(a^2 + 4) = 0$$

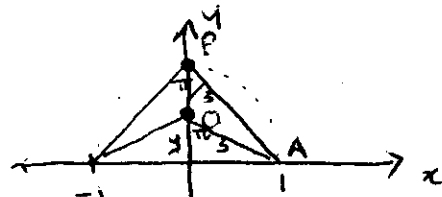
$$\therefore a = 3, b = -2$$

$$a = -3, b = 2$$

(d)

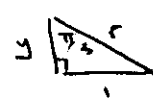


(e)



P lies on circumference

$\therefore$ Centre subtends $\frac{\pi}{3}$ with origin and (1,0)



$$\tan\frac{\pi}{3} = \frac{y}{1}$$

$$y = \frac{1}{\sqrt{3}}$$

$$\therefore \text{centre } (0, \frac{1}{\sqrt{3}})$$

WRONG QUESTION - should have been

Show that $a^2 - b^2 = 5$ and $ab = -6$

$$\sin\frac{\pi}{3} = \frac{1}{r}$$

$$r = \frac{1}{\sin\frac{\pi}{3}} \text{ radius}$$

$$x^2 + \left(y - \frac{1}{\sqrt{3}}\right)^2 = \frac{4}{3}$$

only for 2nd mk.

$$(f) z_2 = iz_1$$

$$z_1 + z_2 = z_1 + iz_1$$

$$= z_1(1+i)$$

$$(z_1 + z_2)^2 = z_1^2(1+i)^2$$

$$= z_1^2(1+2i-1)$$

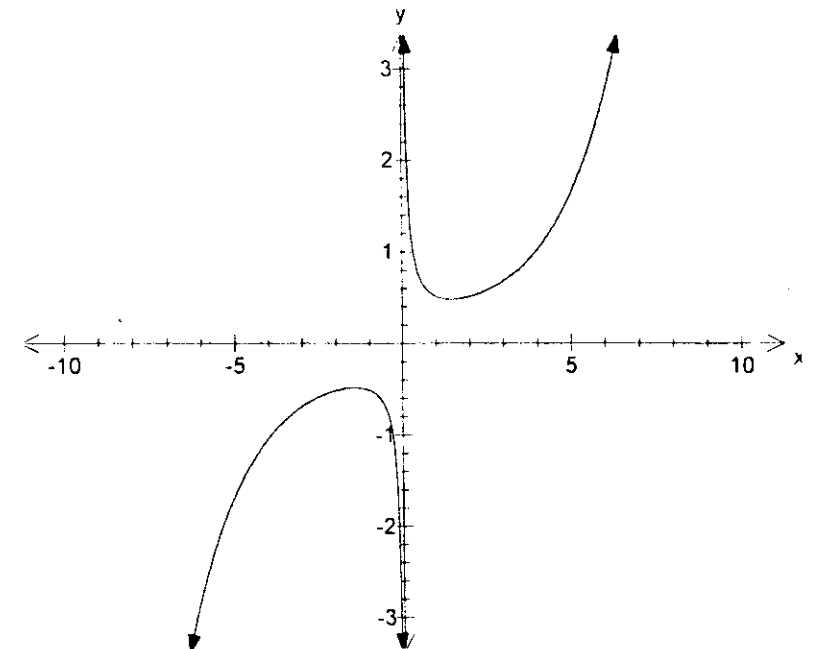
$$= 2iz_1^2$$

$$= 2z_1^2(iz_1)$$

$$= 2z_1 z_2$$

Question 3

$$(a) (i) v = \frac{1}{f(x)}$$



✓ asymptote at $x=0$

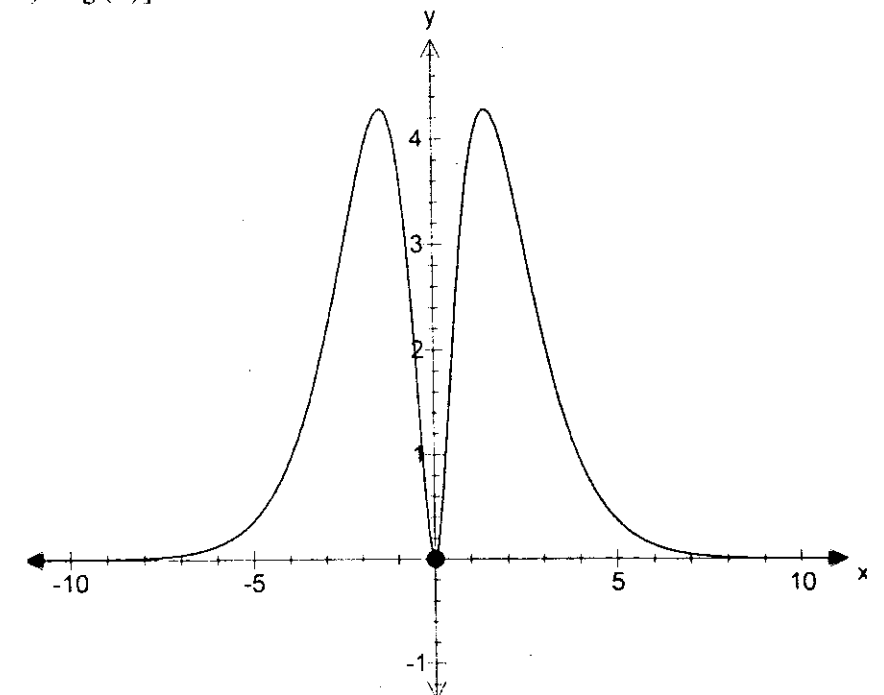
$$(4, 1)$$

$$(-4, -1)$$

$$(1.5, \frac{1}{2})$$

$$(-1.5, -\frac{1}{2})$$

$$(ii) y = [f(x)]^2$$



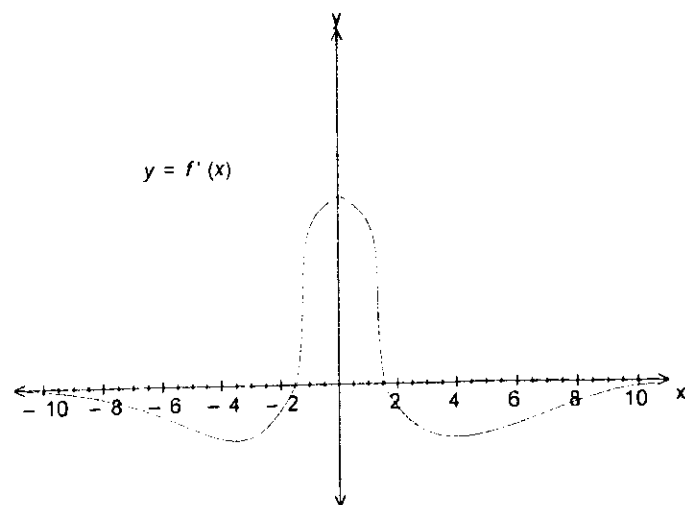
✓ corner at origin

✓ shape.

(max at $x = 4.5, 1.5$)

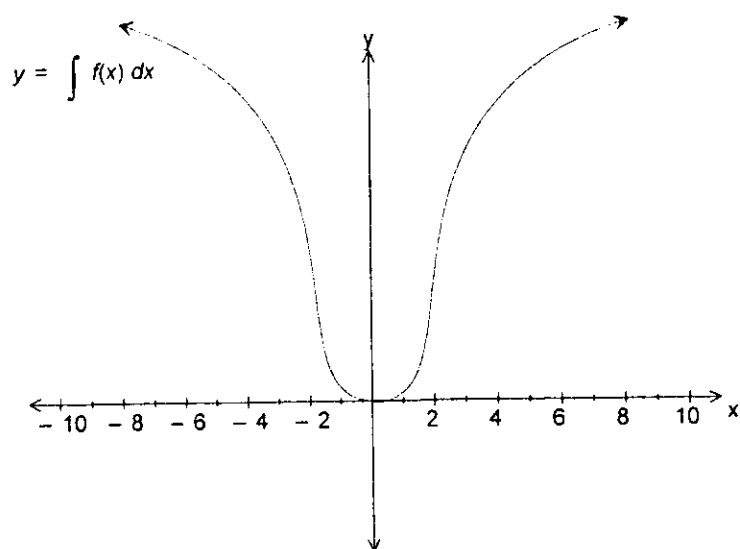
(iii)

- ✓ max tp at $x=0$
- ✓ x intercepts at $x=-1.5, 1.5$

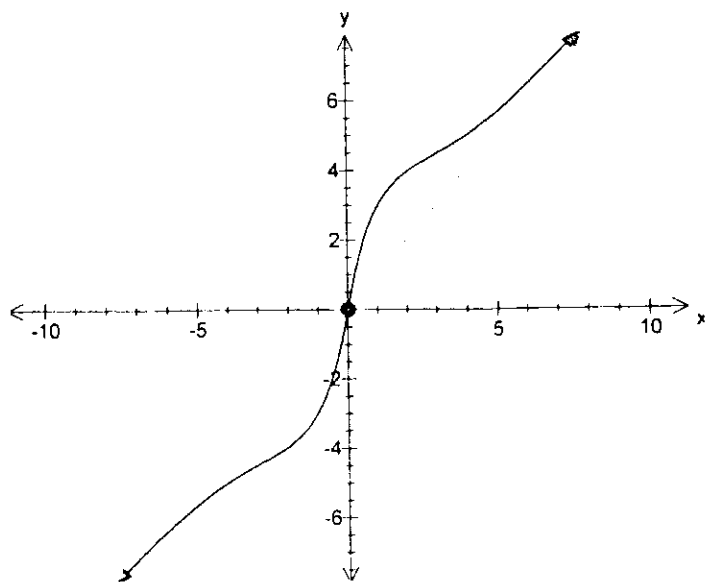


(iv)

- ✓ min t.p at $(0,0)$
- ✓ dec. at dec. rate $x < 0$
- increasing at dec. rate $x > 0$

(v) $y = x + f(x)$

- ✓ thru $(0,0)$
- ✓ symmetric about origin



Qn. 3. (10+)

(b) (i), when $x=a$,

$$y = \pm \sqrt{1-a^2}$$

$$\therefore \text{length of base} = 2\sqrt{1-a^2} \quad \checkmark$$

$$h^2 = \left(\frac{3}{2}\sqrt{1-a^2}\right)^2 - \left(\sqrt{1-a^2}\right)^2$$

$$= \frac{9}{4}(1-a^2) - (1-a^2)$$

$$= \frac{5}{4}(1-a^2)$$

$$\therefore h = \frac{\sqrt{5(1-a^2)}}{2} \quad \checkmark$$

$$\therefore \text{Area} = \frac{1}{2} \times 2\sqrt{1-a^2} \times \frac{\sqrt{5}\sqrt{1-a^2}}{2}$$

$$= \frac{\sqrt{5}}{2}(1-a^2) \text{ units}^2 \quad \checkmark$$

(ii) $\delta v = A \cdot \delta x$ where A is area of isosceles Δ

$$= \frac{\sqrt{5}}{2}(1-x^2) \delta x$$

$$\therefore v = \frac{\sqrt{5}}{2} \int_0^1 (1-x^2) dx \quad \checkmark$$

$$= \frac{\sqrt{5}}{2} \left[x - \frac{x^3}{3} \right]_0^1$$

$$= \frac{\sqrt{5}}{2} \left[\left(1 - \frac{1}{3}\right) - 0 \right]$$

$$= \frac{\sqrt{5}}{3} \text{ units}^3 \quad \checkmark$$

Question 4

(a) Let $m = \frac{1}{x}$

Since $x = p, q, r$ then

$$m = \frac{1}{p}, \frac{1}{q}, \frac{1}{r}$$

Subst. $x = \frac{1}{m}$ into eqn.

$$\frac{1}{m^3} + \frac{4}{m^2} - \frac{3}{m} + 1 = 0 \quad \checkmark$$

$$1 + 4m - 3m^2 + m^3 = 0$$

$\therefore$ the equation is

$$x^3 - 3x^2 + 4x + 1 = 0 \quad \checkmark$$

(b) (i) $F(x) = (x-k)Q(x) \quad \checkmark$

$$F'(x) = Q(x) + (x-k)Q'(x) \quad \checkmark$$

$$\therefore F'(k) = Q(k) \quad \checkmark$$

Since k is a zero of $F'(x)$,

$$Q(k) = 0$$

So, by the factor thm

$(x-k)$ is a factor of $Q(x)$

$$\text{So, } F(x) = (x-k)(x-k)Q^*(x)$$

$$\therefore (x-k)^2 \text{ is a factor of } F(x) \quad \checkmark$$

(ii) $y^{2t} - ty^{t+1} = 1 - ty^{t-1}$

$$y^{2t} - ty^{t+1} + ty^{t-1} - 1 = 0$$

$$\text{Let } P(y) = y^{2t} - ty^{t+1} + ty^{t-1} - 1$$

$$P(1) = 0 \quad \checkmark$$

$$P'(y) = 2ty^{2t-1} - t(t+1)y^t + t(t-1)y^{t-2}$$

$$P'(1) = 0 \quad \checkmark$$

$\therefore$ So, $y=1$ is a zero of $P(y)$ and $P'(y)$
 $\therefore$ it must be a double root
 (root of multiplicity 2) of $P(y)=0$

(d) (i) $(\cos \theta + i \sin \theta)^5 = 1$

$$\cos 5\theta + i \sin 5\theta = 1$$

$$\therefore \cos 5\theta = 1 \quad \checkmark$$

$$5\theta = 0, 2\pi, 4\pi, 6\pi, 8\pi$$

$$\therefore \theta = 0, \frac{2\pi}{5}, \frac{4\pi}{5}, \frac{6\pi}{5}, \frac{8\pi}{5}$$

$$\therefore z = 1, \cos \frac{2\pi}{5}, \cos \frac{4\pi}{5},$$

$$\cos \frac{6\pi}{5}, \cos \frac{8\pi}{5} \quad \checkmark$$

(ii) (a)

$$z^5 - 1 = (z-1)(z - \cos \frac{2\pi}{5})(z - \cos \frac{4\pi}{5})$$

$$(z - \cos \frac{6\pi}{5})(z - \cos \frac{8\pi}{5}) \quad \checkmark$$

(b)

$$z^5 - 1 = (z-1)(z - \cos \frac{2\pi}{5})(z - \cos \frac{4\pi}{5})$$

$$(z - \cos(-\frac{4\pi}{5}))(z - \cos(-\frac{2\pi}{5}))$$

$$\cos(-\frac{4\pi}{5}) = \cos(\frac{4\pi}{5}) + i \sin(-\frac{4\pi}{5})$$

$$= \cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5}$$

$$(\cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5})(\cos \frac{4\pi}{5} - i \sin \frac{4\pi}{5})$$

$$= \cos^2 \frac{4\pi}{5} + \sin^2 \frac{4\pi}{5}$$

$$= 1$$

$$\therefore (z - \cos \frac{4\pi}{5})(z - \cos(-\frac{4\pi}{5})) \quad \checkmark$$

$$= z^2 - 2z \cos \frac{4\pi}{5} + 1$$

$$\therefore z^5 - 1 = (z-1)(z^2 - 2z \cos \frac{2\pi}{5} + 1)$$

$$(z^2 - 2z \cos \frac{4\pi}{5} + 1) \quad \checkmark$$

Qn. 4 (c).

$$z = \frac{1-t^2}{1+t^2} + \frac{2t}{1+t^2} i$$

$$|z| = \sqrt{\left(\frac{1-t^2}{1+t^2}\right)^2 + \left(\frac{2t}{1+t^2}\right)^2} \quad \checkmark$$

$$\text{as } \left\{ \begin{array}{l} z = x+iy \\ |z| = \sqrt{x^2+y^2} \end{array} \right\}$$

$$= \sqrt{\frac{1-2t^2+t^4+4t^2}{(1+t^2)^2}}$$

$$= \sqrt{\frac{1+2t^2+t^4}{(1+t^2)^2}}$$

$$= \sqrt{\frac{(1+t^2)^2}{(1+t^2)^2}} \quad \checkmark$$

$$= 1$$

Question 5

(a) (i) $S(ae, 0)$
 $\therefore$ line thru S has
equation $x = ae$

$\therefore$ At P and Q ,

$$\frac{(ae)^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$y^2 = b^2(e^2 - 1) \quad \checkmark$$

but $b^2 = a^2(e^2 - 1)$

$$e^2 - 1 = \frac{b^2}{a^2}$$

$$\therefore y^2 = b^2 \cdot \frac{b^2}{a^2}$$

$$y^2 = \frac{b^4}{a^2}$$

$$y = \pm \frac{b^2}{a} \quad \checkmark$$

$\therefore P, Q$ have coords.

$$(ae, \pm \frac{b^2}{a})$$

$$\therefore PQ = \frac{2b^2}{a}$$

$$(ii) \frac{2b^2}{a} = 2 \cdot 24 = 48$$

$$\therefore b^2 = 24a \quad \checkmark$$

subst. into $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\frac{9^2}{a^2} - \frac{24^2}{b^2} = 1 \quad \checkmark$$

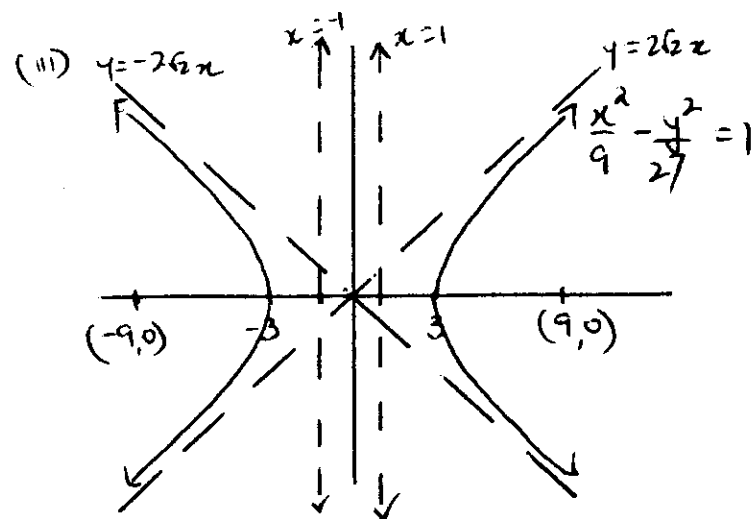
$$a^2 + 24a - 81 = 0$$

$$(a+27)(a-3) = 0$$

$$\therefore a = 3 \quad (\text{not } a = 27 \text{ as } 0 < a < b)$$

$$\text{and } b^2 = 24 \cdot 3 = 72$$

$$b = \sqrt{72} = 6\sqrt{2}$$



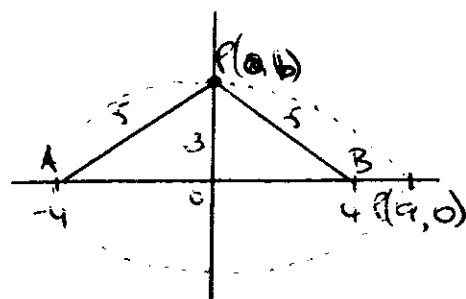
x-intercept $x = \pm 3 \quad \checkmark$

Foci $x = \pm ae = \pm 9 \quad \checkmark$

Directrices $x = \pm \frac{a}{e} = \pm 1 \quad \checkmark$

Asymptotes $y = \pm \frac{b}{a}x$
 $= \pm 2\sqrt{2}x \quad \checkmark$

(b) (i)



$OP = 3$ (by Pythagoras)

$$\therefore b = 3 \quad \checkmark$$

when P is at $(a, 0)$, $a = 5$

subst. into $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$$\therefore \text{eqn. is } \frac{x^2}{25} + \frac{y^2}{9} = 1 \quad \checkmark$$

Qn. 5b)

$$(ii) \frac{x^2}{25} + \frac{y^2}{9} = \frac{5^2 \cos^2 \theta}{25} + \frac{3^2 \sin^2 \theta}{9}$$

$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1.$$

= RHS $\quad \checkmark$

$$x = 5 \cos \theta$$

$$\frac{dx}{d\theta} = -5 \sin \theta$$

$$y = 3 \sin \theta$$

$$\frac{dy}{d\theta} = 3 \cos \theta$$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \cdot \frac{d\theta}{dx}$$

$$= 3 \cos \theta \cdot \frac{-1}{5 \sin \theta} \quad \checkmark$$

when $\theta = \frac{\pi}{6}$,

$$\frac{dy}{dx} = \frac{-3 \cdot \frac{\sqrt{3}}{2}}{5 \cdot \frac{1}{2}}$$

$$= \frac{-3\sqrt{3}}{5} \quad \checkmark$$

$$x = 5 \cdot \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}$$

$$y = 3 \cdot \frac{1}{2} = \frac{3}{2}$$

$$\therefore y - \frac{3}{2} = -\frac{3\sqrt{3}}{5} \left(x - \frac{5\sqrt{3}}{2} \right)$$

or

$$3(3x + 5y - 30) = 0$$

Question 8

(10)

Show that $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$ ✓

$$a^2 + b^2 \geq 2ab \quad (1)$$

$$b^2 + c^2 \geq 2bc \quad (2)$$

$$c^2 + a^2 \geq 2ca \quad (3)$$

① + ② + ③

$$2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

$$a^2 + b^2 + c^2 - ab - bc - ca \geq 0 \quad \checkmark$$

Multiplying b.s. by (a+b+c)

$$(a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca) \geq 0$$

$$\begin{aligned} & a^3 + ab^2 + ac^2 - a^2b - abc - ac^2 \\ & + ba^2 + b^3 + bc^2 - ab^2 - b^2c - abc \\ & + ca^2 + cb^2 + c^3 - abc - bc^2 - ac^2 \geq 0 \end{aligned}$$

$$a^3 + b^3 + c^3 - 3abc \geq 0$$

$$a^3 + b^3 + c^3 \geq 3abc$$

$$\text{let } a^3 = a, b^3 = b, c^3 = c$$

$$\therefore a + b + c \geq 3\sqrt[3]{abc} \quad \checkmark$$

$$\therefore \frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

$$\therefore P(x) = (2x^2 - 5x + 2)Q(x) + R(x)$$

$$\text{Since } \deg P(x) > \deg R(x)$$

$$\deg R(x) < 2$$

$$\text{Let } R(x) = ax + b$$

$$\therefore P(x) = (2x^2 - 5x + 2)Q(x) + ax + b \quad \checkmark$$

$$P\left(\frac{1}{2}\right) = \frac{5}{2} + b = -2 \quad \dots (1)$$

$$P(2) = 2a + b = 1 \quad \dots (2) \quad \checkmark$$

② - ①

$$\frac{3a}{2} = 3$$

$$\therefore a = 2, b = -3$$

$$\therefore R(x) = 2x - 3 \quad \checkmark$$

(c)(i).

$$P(x) = ax^4 + bx^3 + cx^2 + dx + e = 0$$

$$P'(x) = 4ax^3 + 3bx^2 + 2cx + d$$

$$P''(x) = 12ax^2 + 6bx + 2c$$

$$P'''(x) = 24ax + 6b = 0 \quad \checkmark$$

as α is a quadruple root

$$\therefore x = -\frac{6b}{24a} = -\frac{b}{4a} \quad \checkmark$$

$$(ii) \text{ Since } \alpha = -\frac{b}{4a} \text{ is a}$$

quadruple root,

$$P(x) = a(x - \alpha)^4 = ax^4 + bx^3 + cx^2 + dx + e$$

$$\therefore (x - \alpha)^4 = \frac{ax^4 + bx^3 + cx^2 + dx + e}{a} \quad \checkmark$$

$$\text{Subst. } x = 1 \text{ and } \alpha = -\frac{b}{4a}$$

$$\left(1 + \frac{b}{4a}\right)^4 = \frac{a + b + c + d + e}{a} \quad \checkmark$$

Q.8(d)

$$(i) \int_1^e x^{-1} \log x \, dx$$

$$= \frac{1}{2} [\log x]_1^e \quad \checkmark$$

$$= \frac{1}{2} \quad \checkmark$$

$$(ii) \text{ Let } I = \int_1^e x^n \log x \, dx$$

$$\text{let } u = \log x \quad v = x^n$$

$$\frac{du}{dx} = \frac{1}{x} \quad v = \frac{x^{n+1}}{n+1}$$

index

$$\therefore I = \frac{1}{n+1} [x^{n+1} \log x]_1^e - \frac{1}{n+1} \int_1^e \frac{x^{n+1}}{x} \, dx \quad \checkmark$$

$$= \frac{1}{n+1} (e^{n+1}) - \frac{1}{n+1} \int_1^e x^n \, dx$$

$$= \frac{1}{n+1} e^{n+1} - \frac{1}{(n+1)^2} [x^{n+1}]_1^e \quad \checkmark$$

$$= \frac{1}{(n+1)^2} [(n+1)e^{n+1} - (e^{n+1} - 1)]$$

$$= \frac{ne^{n+1} + 1}{(n+1)^2} \quad \checkmark$$